PHYS2500J RC: Electromagnetic Induction

Xu ZHANG

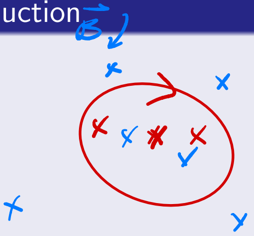
November 17, 2024

Concept Recap

Really, not much to memorize.

Faraday's Law of Electromagnetic Induction

$\epsilon = -\frac{d\Phi}{dt}$ ✓



We have several common situations:

- $\epsilon = -B\frac{dS}{dt}$
- $\epsilon = Blv$
- $\epsilon = -\frac{dB}{dt}S$

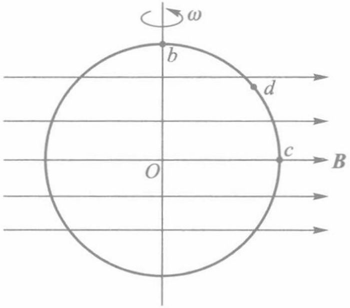
Lenz's Law

The induced current tends to generate a magnetic field that resists the current change of magnetic flux.

Q.0: A Conceptual Question (Skip)

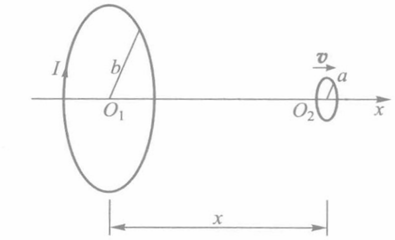
As shown in the figure below, a ring is rotating around an axis which is perpendicular to the magnetic field, with an angular velocity ω. When the ring is at the location shown in the graph, compare:

- Electrical potential of points b & c.
- Electrical potential of points d & c.



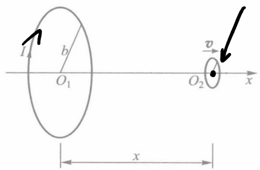
Q.1

As shown in the figure below, two rings are arranged coaxially, with radius b and a ($b >> a$). In the big ring there is a current I. The small ring is moving rightwards with constant velocity $\vec{v}$. Find the induced electromotive force in the small ring when the two rings are of distance x apart. Consider $x >> R$.



As shown in the figure below, two rings are arranged coaxially, with radius b and a ($b >> a$). In the big ring there is a current I. The small ring is moving rightwards with constant velocity $\vec{v}$. Find the induced electromotive force in the small ring when the two rings are of distance x apart. Consider $x >> R$.

$\vec{B} = \text{const}$



Current I (circle):

magnetic field $\vec{B}$ at distance x is:

$$\vec{B}_x = \frac{\mu_0 I b^2}{2(b^2+x^2)^{\frac{3}{2}}}$$

$$\Phi_x = \vec{B}_x \cdot \vec{S}$$

$$= \frac{\mu_0 I b^2}{2(b^2+x^2)^{\frac{3}{2}}} \pi a^2$$

$$\frac{d\Phi_x}{dt} = \frac{\pi \mu_0 I b^2 a^2}{2} \cdot \frac{d}{dt}\left(\frac{1}{(b^2+x^2)^{\frac{3}{2}}}\right)$$

$$\frac{dx}{dt} = v$$

$$v$$

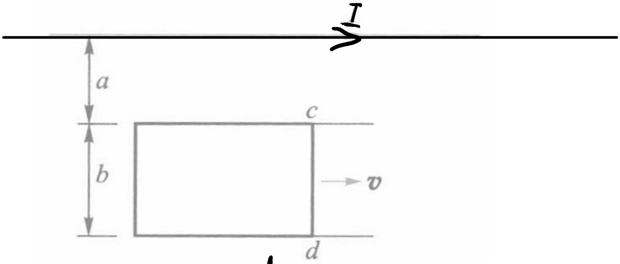
$$= \frac{3\mu_0 \pi I a^2 b^2 x}{2\left(b^2+x^2\right)^{5/2}} \cdot \boxed{\frac{dx}{dt}}$$

$$x \gg b$$

$$= \frac{3\mu_0 \pi I a^2 b^2 \cancel{x}}{2x^{\cancel{5}\,4}} \; v$$

Q.2

As shown in the figure below, Wires form a rectangle beside an infinitely long wire in the same plain, where wire *cd* can move freely. Take the current in the infinite wire to be $I(t) = I_0 e^{\lambda t} (\lambda > 0)$, velocity of wire *cd* to be $\vec{v}$, heading rightwards.



$\Phi = \vec{B} \cdot \vec{S}$

$\frac{dB}{dt} S$ ×

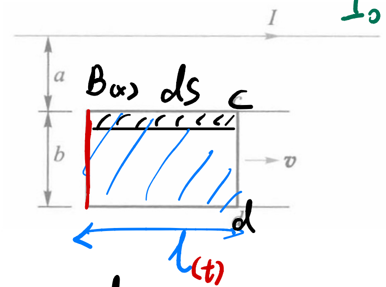
$B \cdot \frac{dS}{dt}$ ×

$\Phi(t)$

$\frac{d\Phi(t)}{dt} = \mathcal{E}$

As shown in the figure below, Wires form a rectangle beside an infinitely long wire in the same plain, where wire *cd* can move freely. Take the current in the infinite wire to be $I(t) = I_0 e^{\lambda t} (\lambda > 0)$, velocity of wire *cd* to be $\vec{v}$, heading rightwards.

$\to I_0 e^{-\lambda t}$



$x : a \to a+b$

$$\Phi_{(t)} = \int_a^{a+b} B_{(x)} \cdot ds_{(x)}$$

$$ds : l\, dx$$

$$B(x) : \frac{\mu_0 I}{2\pi x}$$

$$\int_a^{a+b} \frac{\mu_0 I}{2\pi x} l\, dx = \boxed{\frac{\mu_0 I l}{2\pi} \ln \frac{a+b}{a}}$$

$$\Phi(l) = \frac{\mu_0 I l}{2\pi} \ln \frac{a+b}{a}$$

l is changing over time t

$$l = V \cdot t$$

$$\Phi(t) = \frac{\mu_0 \; Vt}{2\pi} \cdot \left(\ln \frac{a+b}{a}\right) I$$

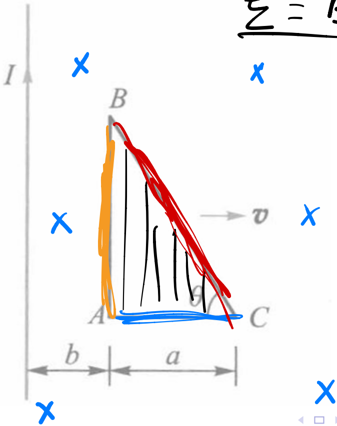
$$= \frac{\mu_0 V t}{2\pi} \ln \frac{a+b}{a} \cdot I_0 e^{-\lambda t}$$

$$\boxed{\mathcal{E} = -\frac{\mu_0 V}{2\pi} I_0 e^{-\lambda t} (1-\lambda t) \ln \frac{a+b}{a}}$$

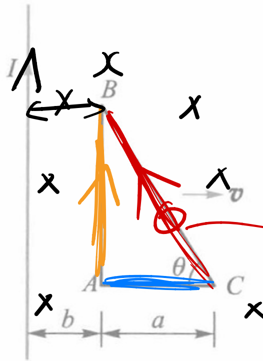
Q.3

An infinite wire is placed in the same plain as a triangular wire ring, as shown below. The straight wire carries a current I, while the triangle moves rightwards with velocity $\vec{v}$. Find the total electromagnetic force in the triangle.

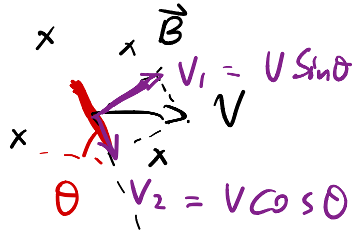
$\underline{\mathcal{E} = Blv}$



An infinite wire is placed in the same plain as a triangular wire ring, as shown below. The straight wire carries a current I, while the triangle moves rightwards with velocity $\vec{v}$. Find the total electromagnetic force in the triangle.



$$\boxed{B_{(x)} = \frac{\mu_0 I}{2\pi x}}$$



① $\mathcal{E}_{AC} = 0$

② $\mathcal{E}_{AB} = B_{(b)} \boxed{a\tan\theta} \cdot V$

$$d\mathcal{E} = B_{(x)} \cdot V\sin\theta \cdot \underline{dl}$$

$$dl = \frac{dx}{\cos\theta}$$

③ $\mathcal{E}_{CB} = \int_a^{a+b} B_{(x)}\, V \sin\theta \frac{dx}{\cos\theta}$

$$\mathcal{E} = \mathcal{E}_{AB} - \mathcal{E}_{CB} = \frac{\mu_0 I V}{2\pi}\left(\frac{a}{b} - \ln\frac{a+b}{b}\right)\tan\theta.$$